

CURRICULUM VITAE

ZAFIR AHMAD NAIK

Ph.D. Genetics & Plant Breeding

Doctoral Researcher, SKUAST-Kashmir

Kashmir, India

Email: zafirahmad198@gmail.com

Phone: +91-7006087884

[LinkedIn](#) | [Google Scholar](#) | [ResearchGate](#) | [Orcid](#) | [Github](#) |

EDUCATION

Ph.D.	Genetics & Plant Breeding Sher-e-Kashmir University of Agricultural Sciences & Technology of Kashmir (SKUAST-Kashmir), 190025 India GPA: 8.31 / 10 Thesis: “Genome-wide Association Studies (GWAS) for Yield and Yield Attributing Traits in Temperate Rice (<i>Oryza sativa</i>).” Advisor: Dr Najeeb ul Rehman Sofi	<i>May 2022 – Sep 2025</i>
MSc	Genetics & Plant Breeding Sher-e-Kashmir University of Agricultural Sciences Technology of Kashmir (SKUAST-Kashmir), 190025 India GPA: 7.92 / 10 Thesis: “Variation for Agronomic and Quantitative Traits in Oats (<i>Avena sativa</i> L.)” Advisor: Dr Noor Ul Saleem Khuroo	<i>Aug 2019 – Mar 2022</i>
B.Sc. Hons	Agricultural Sciences Sher-e-Kashmir University of Agricultural Sciences & Technology of Jammu (SKUAST-Jammu), 180009 India GPA: 6.98 / 10	<i>Aug 2015 – Jul 2019</i>

RESEARCH EXPERIENCE

1	Founder, Data Harvest Labs (https://www.dataharvestlabs.com/)(MSME Registered) May 2020 – Present - Presently working as a Freelance Research Analyst, Data Scientist, and Bioinformatics Consultant through my own MSME-registered research analytics company, Data Harvest Labs, providing analytical and computational research support to scientists, academicians, and research scholars worldwide. - Deliver advanced statistical, bioinformatics, and data science solutions across agricultural, medical, veterinary, biological, and allied life science disciplines, with a focus on reproducible and publication-quality research. - Develop customized analytical pipelines using R, Python, and interactive Shiny applications for GWAS, clinical data analysis, quantitative genetics, experimental
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	design, multivariate statistics, machine learning, and omics data analysis. 4. Collaborate with researchers on study design, data visualization, scientific reporting, and manuscript preparation, while ensuring reproducible workflows and high-quality analytical standards.
2	I have been involved in I've been involved in various research endeavours spanning plant science, contributing my efforts to a broad spectrum of interdisciplinary projects, including: 1. Genome-wide association studies (GWAS), quantitative genetics, and population genomics for dissecting yield, nutritional, and stress-related traits in rice and buckwheat. 2. Genetic diversity, multivariate analysis, and biometrical characterization of rice, oats, common bean, and other crop germplasm to identify superior genotypes for crop improvement. 3. Root phenomics and root system architecture analysis in buckwheat to understand adaptive traits associated with resource-use efficiency and stress resilience. 4. Nutritional genomics and biofortification research, investigating the genetic architecture of nutritional quality, vitamin profiling (Vit-GWAS), and iron–zinc dynamics in cereal and pseudo-cereal crops. 5. Plant molecular biology and stress physiology, focusing on reactive oxygen species (ROS) detoxification mechanisms and molecular responses to abiotic stresses. 6. Plant–microbe interactions and microbiome research, exploring cultivar-specific endophytic microbial communities and their potential roles in plant health and productivity. 7. Secondary metabolite profiling and biochemical characterization of economically important crops using integrated molecular, biochemical, and statistical approaches. 8. Computational biology and bioinformatics, developing reproducible analytical pipelines, custom R scripts, and interactive Shiny applications for genomics, breeding, and multi-omics data analysis. 9. Scientific reviews and translational research covering fungicide resistance, molecular breeding strategies, and emerging genomic approaches for sustainable crop improvement
3	Rural Agriculture Work Experience [R.A.W.E] from SKUAST-J: 6 months (2018) 1. Completed orientation, village attachment, and hands-on training at Research Stations/KVK/Agri Clinics with project report preparation and evaluation. 2. Trainee at Kanwal Foods & Spices EPIP, SIDCO: Gained expertise in food processing, preservation, packaging, storage, and production report analysis.
4	International Intern (Bioinformatics & Genomics) Nextgenhelper, New Delhi (Jan 22 – Feb 22, 2026): Successfully completed a rigorous, one-month international training internship focused on "Bioinformatics, Genomics, and Biostatistics (Basics to Advanced)". During this intensive program, I gained comprehensive theoretical and practical knowledge bridging foundational to advanced methodologies in genomic data analysis and statistical processing

PUBLICATIONS

Scopus Indexed Journals

1. **Naik, Z. A.**, M., Mansoor, M., Sofi, P. A., Dar, Z. A., Mehvish, S., & Khuroo, N. U. S. (2025). Genetic diversity and yield-biochemical trait correlations in oats from the Northwestern Himalayas. *Genetic Resources and Crop Evolution*, 1-15.
2. Naik, M., & **Naik, Z. A.** (2025). Enhancing plant resilience: mechanisms and genetic strategies for detoxification of Reactive oxygen species. *Proceedings of the Indian National Science Academy*, 1-7.
3. Singh, D., Sudan, J., Zaffar, A., **Naik, Z.**, Bhat, M. A., Shikari, A. B., ... & Zargar, S. M. (2024). Comparative root phenotyping unveils key insights into the root system architecture of buckwheat (*Fagopyrum* sp.) species, a valuable crop of western Himalayan Kashmir. *Discover Agriculture*, 2(1), 88.
4. Sudan, J., Urwat, U., Farooq, A., Pakhtoon, M. M., Zaffar, A., **Naik, Z. A.**, ... &

Zargar, S. M. (2023). Explicating genetic architecture governing nutritional quality in pigmented rice. *PeerJ*, 11, e15901.

5. Verma, A., Sudan, J., Jose, R. C., Bagri, J., **Naik, Z. A.**, Sofi, N. R., ... & Zargar, S. M. (2025). Unveiling molecular mechanisms of iron and zinc dynamics in rice. *Plant Science*, 357, 112543.
6. Jan, S., Ashraf Bhat, M., Kumar, A., Altaf Wani, M., Ahmad Bhat, F., Kanth, R. H., ... & **Naik, Z. A.** (2025). Plants Response to Different Abiotic Stresses. *Journal of Crop Health*, 77(5), 1-27.
7. Manzoor, M., Hami, A., Sudan, J., Dall' Acqua, S., Bhat, B., Zafar, A., **Naik, Z.A.**, ... & Zargar, S. M. (2025). Vitamin-B profiling and Vit-GWAS in buckwheat (*Fagopyrum* spp.): a first report. *Physiology and Molecular Biology of Plants*.
8. Sofi, N. R., Dar, M. S., Mir, S., Shikari, A. B., Bharadwaj, N., Aditya, J. P., ... & **Naik, Z.** (2026). Multi-Environment Analysis of Rice (*Oryza Sativa* L.) Genotypes Under Indian Hill Ecologies. *Indian Journal of Genetics and Plant Breeding*, 1-12.

UGC CARE / NAAS / Google Scholar Indexed

9. **Naik, Z. A.**, Verma, A., Sudan, J., Shikari, A. B., Bhat, F. A., Wani, F. J., ... & Zargar, S. M. (2025). Multivariate Analysis of Morphological and Biochemical Traits Revealed Some Promising Pigmented Rice Genotypes of North Western Himalayas for Future Breeding Program. *Indian Journal of Plant Genetic Resources*, 38(02), 227-231.
10. Naik, M., **Naik, Z.**, Mehraj, S., Malla, I. A., Bhat, J. I. A., Bhat, Z., ... & Masoodi, K. (2025). Endophytic Microbiota of Diverse Apple Cultivars Reveals Cultivar-specific Microbial Associations. *Journal of Advances in Biology & Biotechnology*, 28(4), 808-818.

Non-Indexed / Other Publications

11. Wani, S. Q., Wani, T. A., Rasool, T., Muneer, K., Mansoor, T., **Naik, Z. A.**, Mushtaq, S., Malik, S., & Muzaffar, M. (2023). Evolutionary context of fungicide resistance: A comprehensive review. *The Pharma Innovation International Journal*, 12(6), 292–296.
12. Jan, U., Afroza, B., Rashid, R., Sofi, P. A., **Naik, Z. A.**, & Manzoor, I. (2025). Performance evaluation and diversity analysis in common bean (*Phaseolus vulgaris* L.) germplasm. *Journal of Community Mobilization and Sustainable Development*, 20(1), 63–68.
13. Jan, H., Kumar, A., Mapnoo, N. L. A., Verma, L., **Naik, Z.**, Sofi, N. R., Shikari, A. B., Wani, S. H., Mishra, S., Rasheed, Z., Jeelani, F., Ganai, S. A., Lawson, L., Akhter, A., & Khan, G. H. (2025). Unravelling the genetic variability in rice germplasm under temperate conditions of Kashmir valley. *Agriculture Association of Textile Chemical and Critical Reviews Journal*, 2025(1), 171–174. <https://aatcc.peerjournals.net/>
14. Pakhtoon, M. M., Agrawal, A., Sudan, J., Mahajan, R., **Naik, Z.**, Manzoor, M., Sofi, P. A., & Zargar, S. M. (2025). Evaluation of secondary metabolite profiles in common bean (*Phaseolus vulgaris* L.) genotypes from the Western Himalayan Kashmir. *SKUAST Journal of Research*, 27(2),

Book Chapters

1. **Naik, Z. A.**, Mansoor, M., Verma, A., Naik, M., Hajini, M. A., Abeer, Sofi, P. A., Zargar, S. M., & Sofi, N. U. R. (2026). Multi-omics approach for drought resistance in rice. Book title: Rice Omics and Precision Breeding: Advances in Sustainable Production (Hard ISBN: 9781779648778 E-Book ISBN: 978-1-77964-878-5 Pages: Est 266 pp w index). Publisher: Apple Academic Press
2. **Naik, Z. A.**, Madeeha, Bhat, S. N., Dar, M. S., & Wani, S. H. (2025). Genomic methods for improving abiotic stress tolerance in crops. Fruit breeding: A genomic approach (pp. 241–265). Agrobios Research. ISBN: 978-93-94380-00-0

3. Madeeha, **Naik, Z. A.**, & Jan, U. (2024). Embryo rescue in horticultural crops, Theory and guide to plant tissue culture in horticultural crops (pp. 67–83). Narendra Publishing House.

TECHNICAL EXPERTISE

1. Molecular Biology Techniques, Genotyping, Quantitative genetics, Plant Breeding, Experimental design and statistical analysis, Bioinformatics tools, Scientific communication, Scientific writing.
2. Phenomics and Genomics: Expertise in molecular breeding techniques, NGS data analysis, QTL/GWAS, genomic selection, designing field trials, managing phenotyping pipelines, and analyzing high-throughput phenotypic and genotypic data.

MAJOR SKILLS

1. Data analysis, Proficient in R, Basic Genomics Data Science, Statistical analysis, Statistical Modeling, Clinical Data Analysis, Machine Learning, EDA, Data Visualization, Experimental Design, High-Dimensional Data, Biostatics • Descriptive and inferential statistics for biomedical data, Hypothesis testing, Genomics analysis, MLM/GLM Testing
2. **Programming Languages:**
 - a) **R** (Proficient – statistical modelling, data visualization, machine learning,
 - b) **R Packages:** tidyverse, dplyr, data.table, ggplot2, Shiny, caret, lme4, randomForest, glmnet, factoextra, plotly, tidyr)
 - c) **Python** (Intermediate – basic data manipulation, scripting)
3. **Laboratory Techniques:** DNA Isolation, PCR, qPCR, Gel Electrophoresis, SDS PAGE, breeding techniques, NGS analysis, GWAS analysis of traits, QTL mapping.
4. **Software:** TASSEL, GIT, BASH, GenAlex, DARWIN, MEGA, PowerMarker, JASP, Microsoft Office, Canva.

OACHIEVEMENTS

1. **Patents:** Antifungal composition for use as biocontrol agent against apple scab and method of preparation thereof | Status: Filed | CBR No. 30490 | Application No. 202511055868
2. Reviewed for various reputed journals like BMC Biology

Projects & Software Development

Project / Package Name	Key Features & Technical Implementations	Core Tech Stack
PutativeQTLmapper	Designed and developed an interactive Shiny application optimized for genome-wide association studies (GWAS). Streamlines data translation by dynamically visualizing and mapping putative Quantitative Trait Loci (QTLs) directly from complex GWAS summary statistics.	R, Shiny, ggplot2, plotly, Bioinformatics
met-stability-analyzer	Engineered an automated, end-to-end Shiny pipeline designed for Multi-Environment Trial (MET)	R, Shiny, dplyr, ggplot2, DT,

	agricultural data analysis. Features single-click execution for Critical Difference (CD), Environment ANOVA, Genotype-by-Environment ($G \times E$) ANOVA, and Eberhart-Russell stability modeling on multi-trait datasets.	Agronomy Analytics
ShinyPCA	Built an intuitive, interactive Principal Component Analysis (PCA) dashboard tailored for high-dimensional biological and multivariate datasets. Enables real-time variance analysis, exploratory data visualization, and automated cluster identification.	R, Shiny, Biostatistics, Data Visualization
E_R Stability	Engineered an automated, single-click coding pipeline implementing Eberhart and Russell's regression model for Genotype-by-Environment ($G \times E$) interaction analysis. Consolidates multi-step statistical computation to output comprehensive stability parameters instantaneously.	R, metan, Linear Regression, Agronomy Analytics

POSTER/ORAL PRESENTATIONS

S.No.	Title of Presentation	Conference theme	organized by	Year
1	Variation for agronomic and quantitative traits in exotic oats (<i>Avena sativa</i> L.) [Poster presentation].	Renaissance of hill agriculture through advanced genetics and crop breeding interventions for attaining food and nutrition security under climate change scenario.	Indian Society of Genetics & Plant Breeding (ISGPB) at SKUAST-Kashmir	2022
2	Unravelling the morphological and nutritional diversity in pigmented rices of Western Himalayan region [Oral presentation]	Biodiversity Awareness to Meet Future Requirements	Department of Botany, University of Kashmir (Sponsored by NASI)	2024

Workshops, Trainings & Professional Certifications

Date	Title / Program Name	Role / Nature of Activity	Organizing Institution / Issuing Authority
July 14–15, 2025	Two Days International Workshop on "OMICS for Crop Improvement"	Participant	Proteomics Lab, Division of Plant Biotechnology, SKUAST-Kashmir (in association with Neuberg Diagnostics)
Oct 23–24, 2024	Two-Days Students' Meet on "Biodiversity Awareness to Meet Future Requirements"	Presenter (Oral/Poster)	Department of Botany, University of Kashmir (Sponsored by NASI)
July 22–	Five Days International	Organizing	Proteomics Lab, Division of Plant

26, 2024	Workshop on "OMICS in Plant Science"	Committee Member	Biotechnology, SKUAST-Kashmir
Mar 8–17, 2023	10 Days Hands-on Training on "Root scanning, non-destructive biochemical and proteomics approaches for abiotic stress tolerance"	Participant	Faculty of Agriculture, Wadura, Sopore, SKUAST-Kashmir
Sept 10–12, 2022	National Symposium on "Renaissance of Hill Agriculture through Advanced Genetics and Crop Breeding Interventions..."	Participant / Presenter (Topic: <i>Variation for agronomic and [...] in oats</i>)	Indian Society of Genetics & Plant Breeding (ISGPB) at SKUAST-Kashmir
Apr 5–9, 2022	5 Days Online Workshop on "Ecological Entrepreneurship: A Foundation Course"	Participant	IDP-NAHEP SKUAST-Kashmir (in association with Sagg Eco-Village Academy)
Mar 27 – Apr 2, 2022	7 Days Training Programme on "Mushroom Production Technology" (under STRY / SMAE / NMAET schemes)	Participant	Directorate of Extension / SAMETI-K, SKUAST-Kashmir (Ministry of Agriculture & Farmers Welfare, GoI)
May 27–29, 2020	3 Days Online Workshop on "Entrepreneurship Sensitization Drive"	Participant	Innovation Cell, IDP-NAHEP, SKUAST-Kashmir
Jan 12–21, 2020	10 Days Skill Development Training on "In-vitro Production of Saffron Cormlets"	Participant	Genome Engineering & Societal Biotechnology Lab, Division of Plant Biotechnology, SKUAST-Kashmir
Aug 3–10, 2020	Webinar Series / Training on "Rejigging the Concept of 'Genetics' with germaneness to translational research"	Participant	IDP-NAHEP, SKUAST-Kashmir
June 2019	Industrial Training / Quality Flowchart Learning	Industrial Trainee	Kanwal Foods & Beverages, Jammu & Kashmir
Mar 5, 2016	Workshop on "Digital India" (Educate Engage Empower)	Participant	National Service Scheme (NSS), Central University of Jammu (in collaboration with NeGD, DeitY, GoI)
Feb 24–28, 2012	5-Day INSPIRE Camp-2012	Participant	Centre of Research for Development (CORD), University of Kashmir (Sponsored by DST, GoI)

LANGUAGES

English: Nice listener, good speaker, advanced reading and writing

Urdu: Excellent listener and speaker, advanced reading and writing

Hindi: Excellent listener and speaker, weak reading and writing

Kashmiri: Excellent listener and speaker, weak reading and writing
Arabic: Elementary

REFERENCES

PROF. MOHD AMIN BEG

King Abdulaziz University, Jeddah, Kingdom of Saudi Arabia

Cell: +966-552822451; E-mail: mabeg51@gmail.com

Nationality: United States of America

Degrees: D.V.M., M.S., Ph.D.

Specialization: Veterinary Reproduction, Gynecology, & Obstetrics

Professor Reproductive Biology

King Fahd Medical Research Centre

King Abdulaziz University

PO Box 80216, Jeddah 21589

Kingdom of Saudi Arabia.

Mobile: +966-552822451

Tel: +1-608-245-3948 (USA)

Dr. Khalid Z Masoodi

Associate Professor, Biotechnology, SKUAST-K

Postdoc (UPMC, PA, USA)

masoodikz@skuastkashmir.ac.in

+91-9797439263

Dr Najeeb UK Rehman Sofi

Associate Director and Professor Genetics and Plant Breeding,

Mountain Research centre for field crops, Khudwani, India

najeeb_sofi@rediffmail.com

+91-7006847501

PERSONAL INFORMATION

Name: Dr Zafir Ahmad Naik

Father's Name: Manzoor Ul Haq Naik

Mother's Name: Zahida Parveen

Sex: Male

Marital Status: Un-married

Nationality: Indian

Language Known: English, Hindi, Kashmiri, Urdu.

Permanent Address: Korel, Kulgam, Jammu and Kashmir, 192233 India.

Corresponding Address: Budshahnagar, Natipora, Srinagar Jammu and Kashmir, 190015 India.

DECLARATION

I hereby declare that all the information provided by me is true to the best of my knowledge.

Place: J&K, India

Date: 2 Feb 2026



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